

Designing Scientific Figures as Visual Arguments

From scientific claims to coherent and trustworthy visual evidence

Information Visualization · Lecture 7

A polished chart can still fail as evidence

Claim: Method A consistently outperforms Method B.

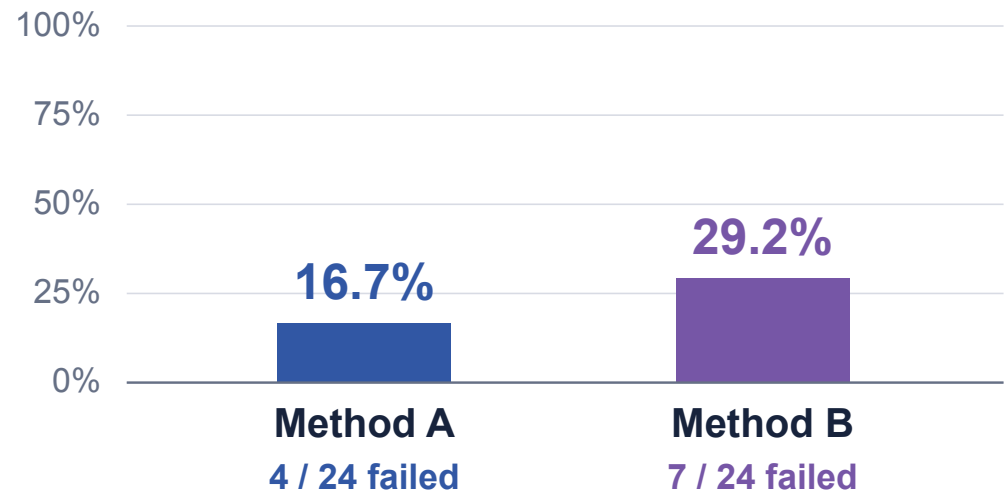
Test scores for A and B: 3 conditions · **8 runs** per method in each condition. (higher score is better.)

Condition	Method A	Method B
Baseline No added noise	77, 78, 79, 80 80, 81, 82, 83	58, 66, 74, 78 82, 86, 96, 100
Mild noise Low noise	73, 74, 75, 76 76, 77, 78, 79	52, 60, 68, 72 76, 80, 90, 94
Strong noise High noise	52, 58, 64, 68 70, 74, 78, 80	58, 66, 70, 72 76, 78, 84, 88

Overall failure fraction

All 3 conditions · 24 runs per method

Failure: score <70 · Lower is better



Method A consistently outperforms Method B.

A visible difference is not yet an inspectable scientific argument.

Scientific evidence figures make claims inspectable

A **scientific evidence figure** organizes visual evidence so readers can evaluate a scientific claim.

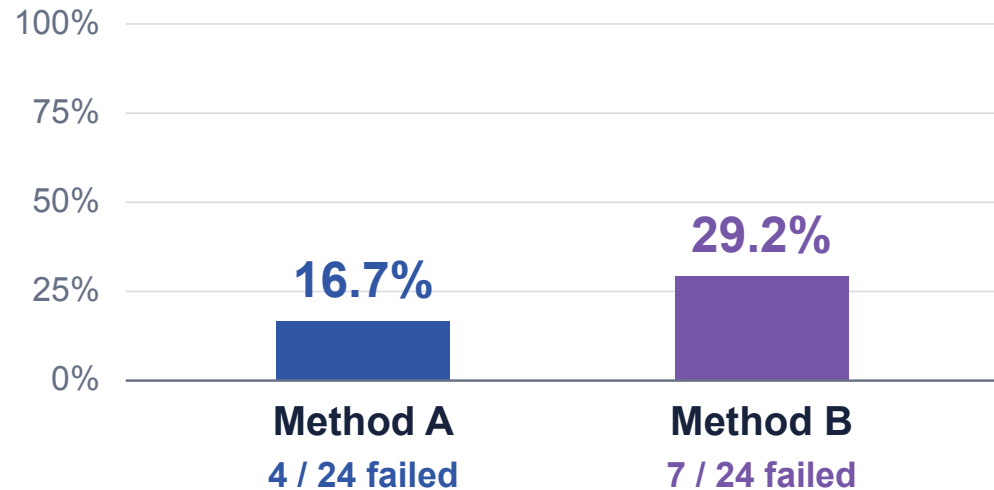
- It makes the relevant comparison visible.
 - It preserves enough context to judge the evidence.
-
- “Visual argument” is shorthand for an argument that remains **inspectable**.
 - Other scientific figures may explain methods, mechanisms, or study designs.

Making the consistency inspectable

Overall failure fraction

All 3 conditions · 24 runs per method

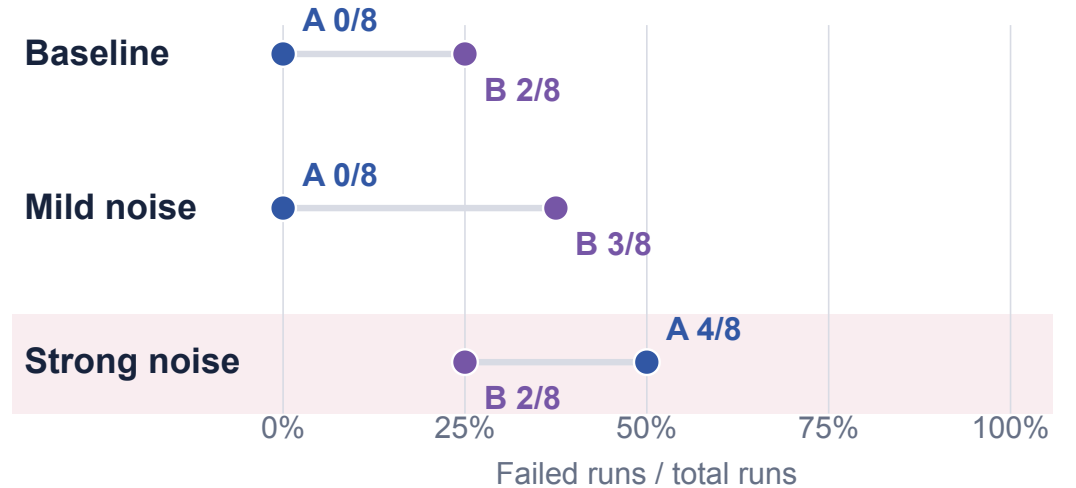
Failure: score <70 · Lower is better



A lower overall failure fraction can hide exceptions in individual conditions.

Observed failures across conditions

Failure: score <70 · 8 runs per method / condition



Showing each condition makes consistency inspectable.

Agenda

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- | | | |
|----|--------------------------------------|--|
| 01 | Claim to comparison | Translate a scientific claim into the judgments readers must make before choosing a chart. |
| 02 | One view or multiple views | Check which judgments one view can support and when another view is needed. |
| 03 | Designing a multi-view figure | Guide reading order, choose useful panels, and coordinate views for comparison. |
| 04 | Integrity | Preserve observations, uncertainty, variability, and analytical context needed for justified interpretation. |
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PART 01

From claim to comparison

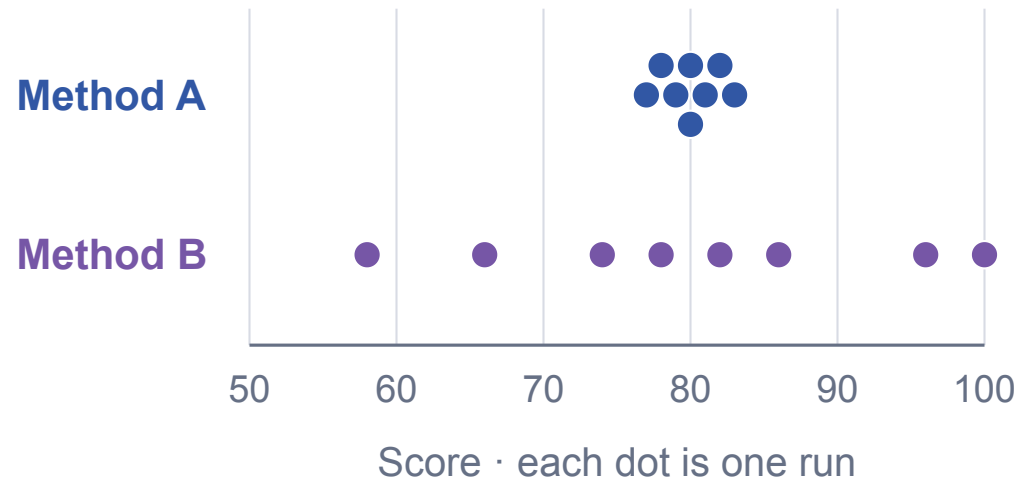
What must readers judge before we choose a chart?

A claim does not determine a chart

Candidate claim: Method A is more reliable than Method B.

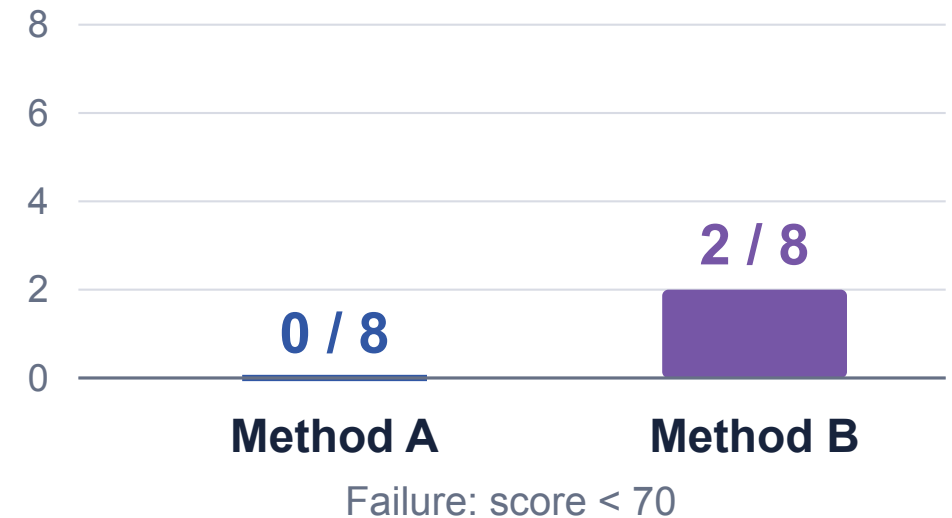
1 · Lower variation

Same 8 baseline runs per method.



2 · Fewer failures

Same 8 baseline runs per method.



Define what “reliable” means before choosing how to show it.

Reader judgments determine what the chart must show

A **scientific claim** states a proposed conclusion. A **reader judgment** is the comparison or assessment needed to evaluate it.

- The judgment guides chart choice
- More than one chart may support the judgment
- Choose a chart whose **encodings make the required judgment easy**

Candidate claim	Reader must judge	Possible chart choice
A is more reliable Meaning: lower variation	How widely do scores vary across runs?	Strip plot of run scores, grouped by method
A is more reliable Meaning: fewer failures	What fraction of runs fail the stated rule?	Bar chart comparing failed / total runs
X and Y are associated	Direction, and strength	Scatterplot of paired X and Y values

Derive reader judgments from the claim

Turn the claim into **questions** about difference, consistency, exceptions, and uncertainty.

Candidate claim: A fails less often than B across the tested noise conditions.

Part of the claim	Judgment needed to evaluate it
“less often than B”	Direction: Is A’s failure fraction lower than B’s?
“less often”	Magnitude: How large is the difference?
“across the tested noise conditions”	Consistency and exceptions: Does the ordering hold in every tested condition? Where does it reverse?
The claim as a whole	Uncertainty: How precisely do these runs estimate failure probabilities?

Turn vague descriptions into measurable criteria

Specify **the quantity, calculation, reference, and scope** so readers know exactly what to compare. Any threshold should be justified by the task.

Example: Define “more reliable” as fewer failures under a stated rule for "A is more reliable than B."

Make explicit	Concrete definition for this example
Quantity + unit	Failure fraction, reported as a percentage; lower means fewer failures.
Calculation + sample	Failed runs ÷ total runs; 8 runs per method / condition.
Decision rule	A run fails if its score is below 70 on a 0–100 score scale.
Comparison + scope	A versus B within each condition: baseline, mild noise, and strong noise.

More criteria can make a figure more informative; **fewer criteria** can keep it simpler.
Choose the criteria needed for the intended reader judgment.

Apply the decision rule to individual runs



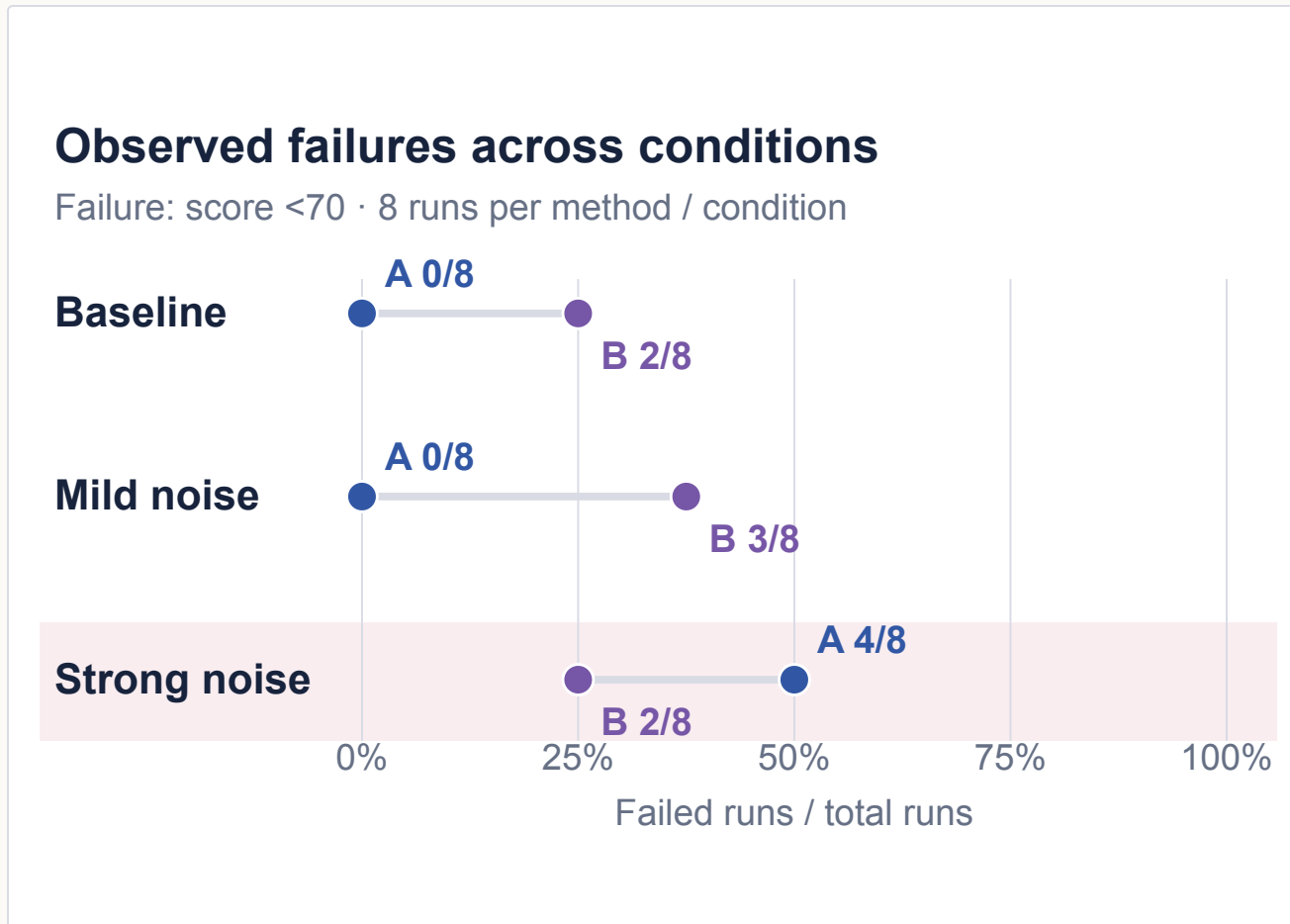
Implemented in the chart

- **Decision rule:** the shaded region identifies scores below 70 as failures.

Not yet shown

- **Quantity + unit:** failure fraction (%) is not plotted; positions show scores.
- **Calculation + sample:** 8 runs are shown per method, but failed ÷ total is not summarized.
- **Comparison + scope:** A and B are compared only at baseline; both noise conditions are missing.

Plot failure fractions across all conditions



Implemented: all four criteria

- **Quantity + unit:** failure fraction (%) on a common horizontal axis.
- **Calculation + sample:** failed / 8 labels accompany each estimate.
- **Decision rule:** score <70 defines failures, as stated above the plot.
- **Comparison + scope:** A versus B in all three conditions.

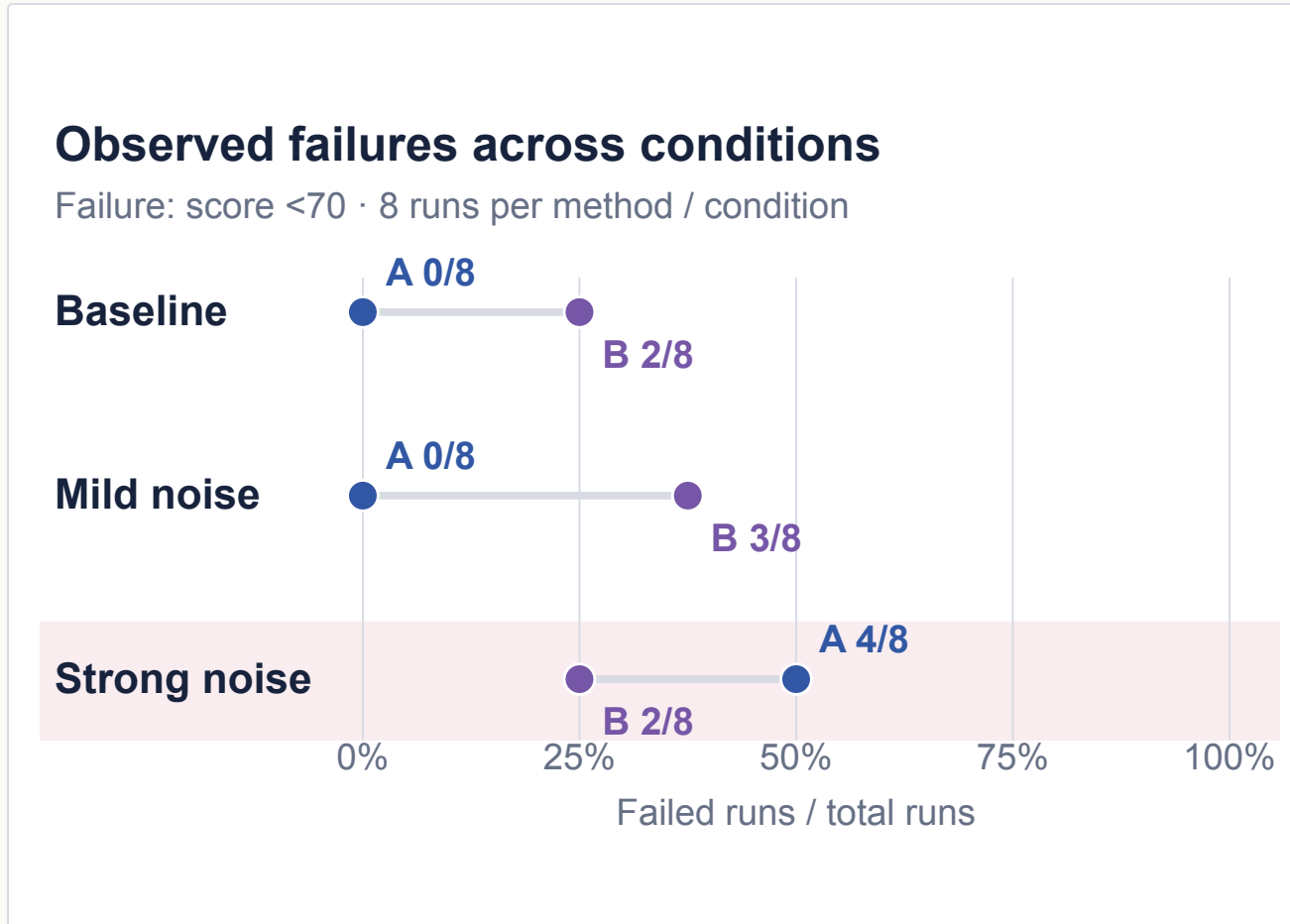
A has fewer observed failures at baseline and under mild noise; the ordering reverses under strong noise.

PART 02

From one view to multiple views

Can one view support the required judgments?

One view can support several reader judgments



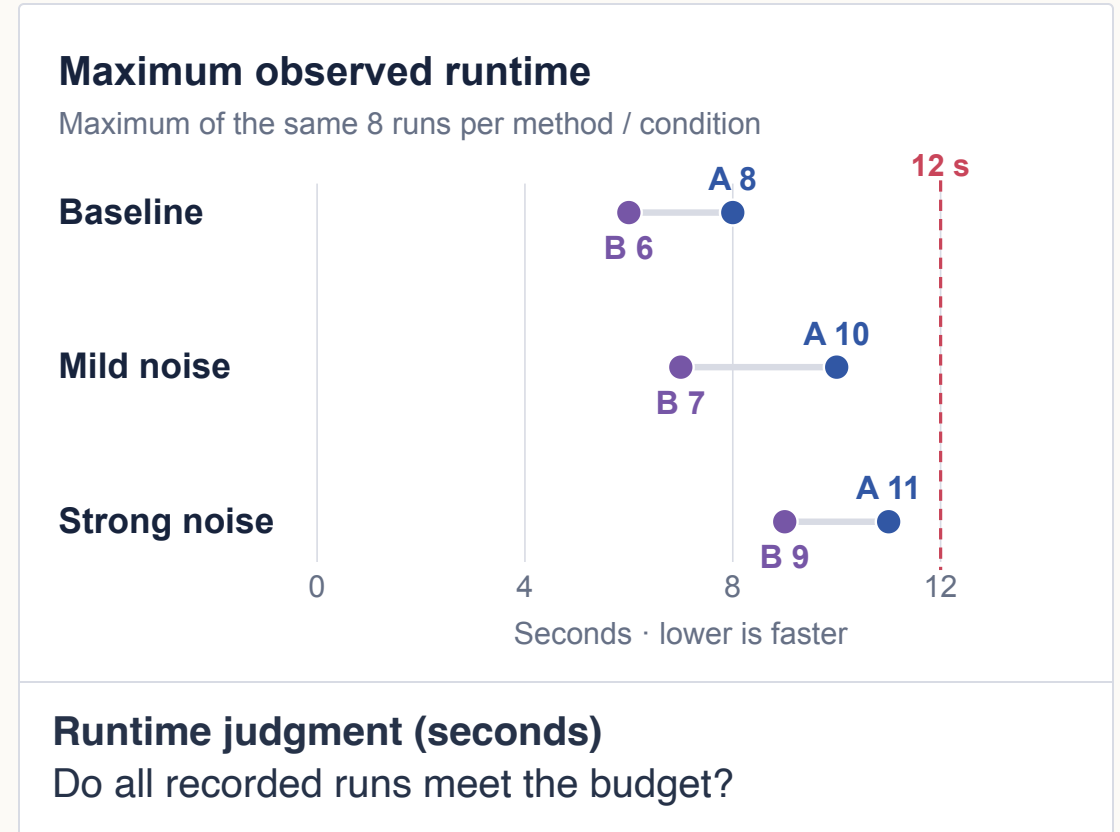
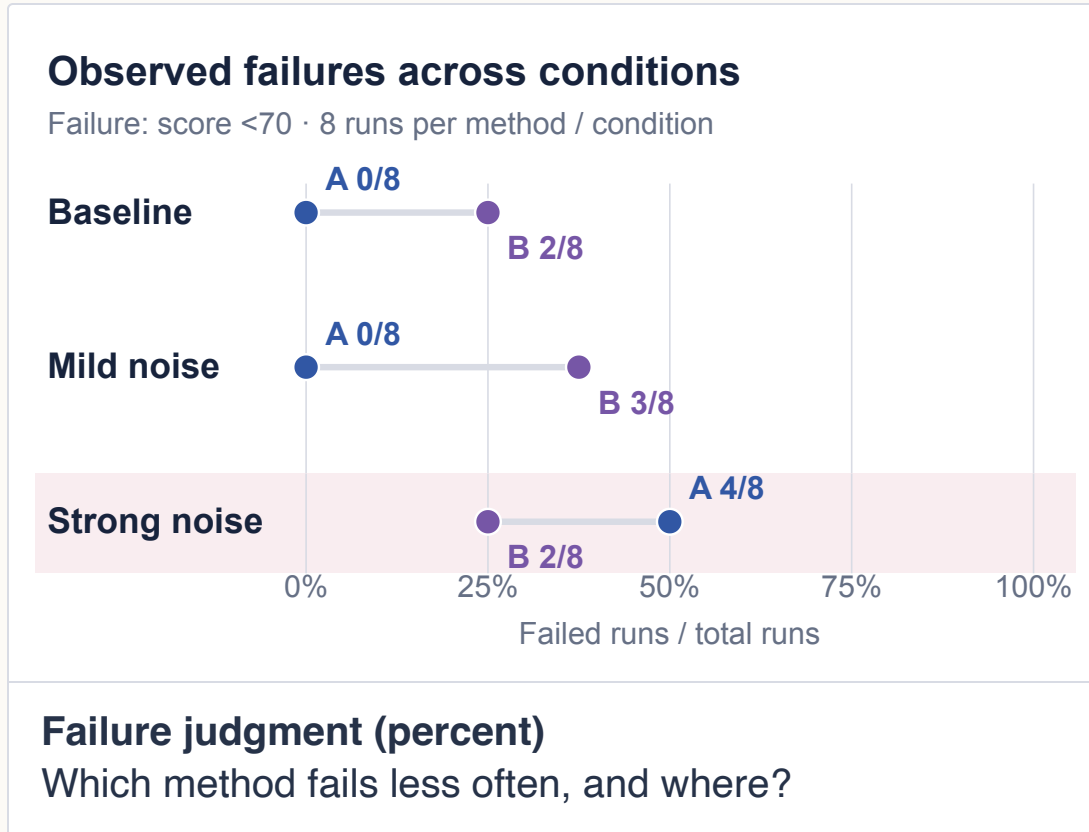
Readers can judge

- **Direction:** which method fails less often in each condition?
- **Magnitude:** how large is each A/B difference?
- **Consistency and exceptions:** does the ordering hold across conditions, and where does it reverse?

Several judgments can share the same marks and scale.

A new judgment can require another view

Additional requirement: each run must finish within **12 seconds**.



A failure fraction cannot show whether a run meets a time budget.

PART 03

Designing a multi-view figure

How can readers follow one argument across several panels?

Three questions guide a multi-view figure

A **multi-view figure** combines related views so readers can inspect different aspects of the scientific question.

For this example, the **candidate claim** is: “A fails less often than B across the tested noise conditions, and every recorded run of both methods finishes within **12 seconds**.”

1. What should readers see first, then next?

First compare failures across conditions; then check whether both methods meet the 12-second limit.

2. Which views should stay?

Keep views that add a required judgment; combine duplicated evidence.

3. How can readers compare across views?

Coordinate views for comparison through alignment, shared encodings, and suitable scales.

A reading path guides where to start and what comes next

A **reading path** is the intended sequence in which readers encounter and connect the views in a figure.

- **Visual hierarchy:** the relative prominence of elements, created by differences in size, position, and contrast.
- **Entry point:** a larger view or a stronger heading can signal where to begin.
- **Sequence:** spatial order and panel labels suggest which view to inspect next.

In our example, readers first compare **failures across conditions**, then check the **runtime constraint**.

Common reading paths use rows, columns, or a grid

Left to right



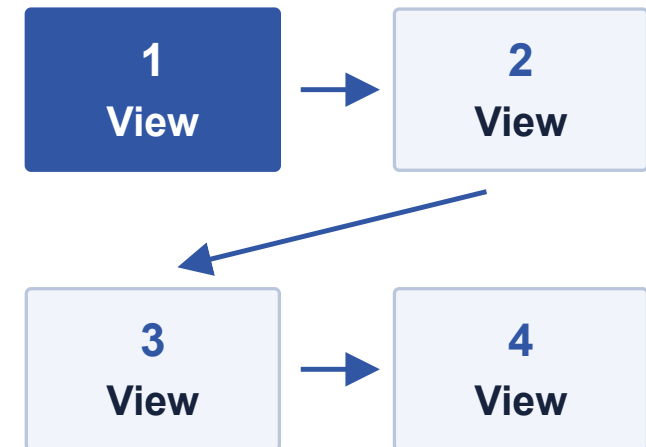
Size emphasizes failures.
Read across the row.

Top to bottom



Bold type emphasizes failures.
Read down the column.

Row by row (Z path)



Contrast marks the entry point.
Finish one row, then the next.

Visual hierarchy signals where to start; position, panel labels, and arrows indicate what comes next.

Three principles determine which views deserve space

- **Priority:** Give the most prominent position and space to the evidence most important to the claim.
Example: prioritize failures across conditions.
- **Complementarity:** A supporting view should enable a relevant judgment that the main view cannot support.
Example: add runtime to check the 12-second limit.
- **Necessity:** Keep a view when removing it would make a required judgment harder or impossible.
Example: do individual scores add a required judgment?

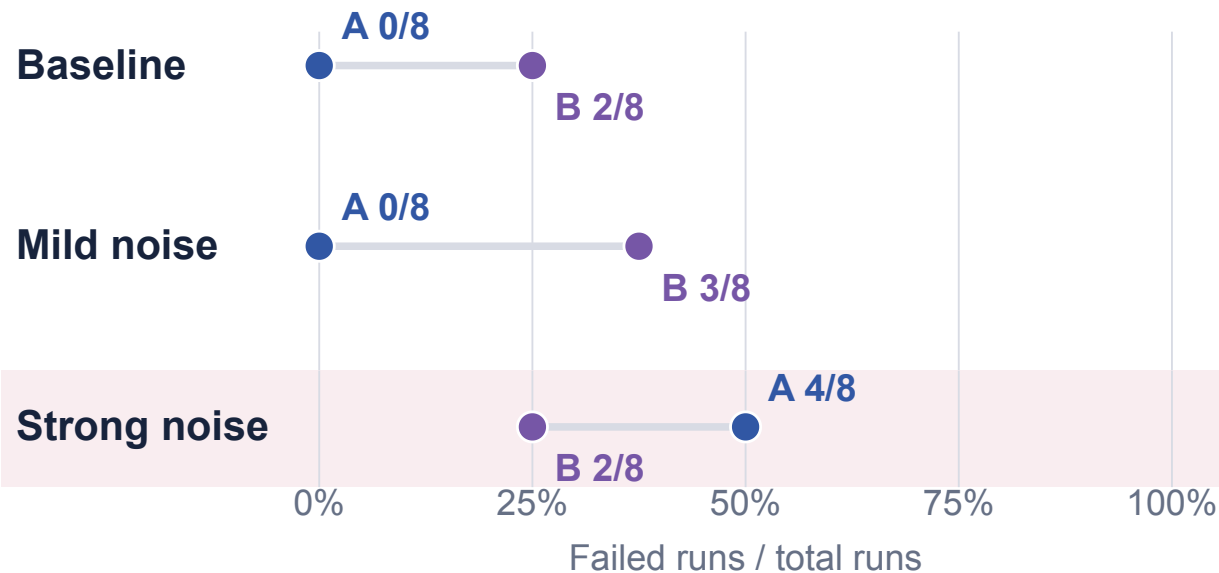
Apply in order: set the priority → add complementary views → test each view's necessity.

Priority makes the reliability comparison the entry point

The layout follows a familiar **left-to-right reading path**: start with the main reliability comparison, then check the additional runtime constraint.

Observed failures across conditions

Failure: score <70 · 8 runs per method / condition

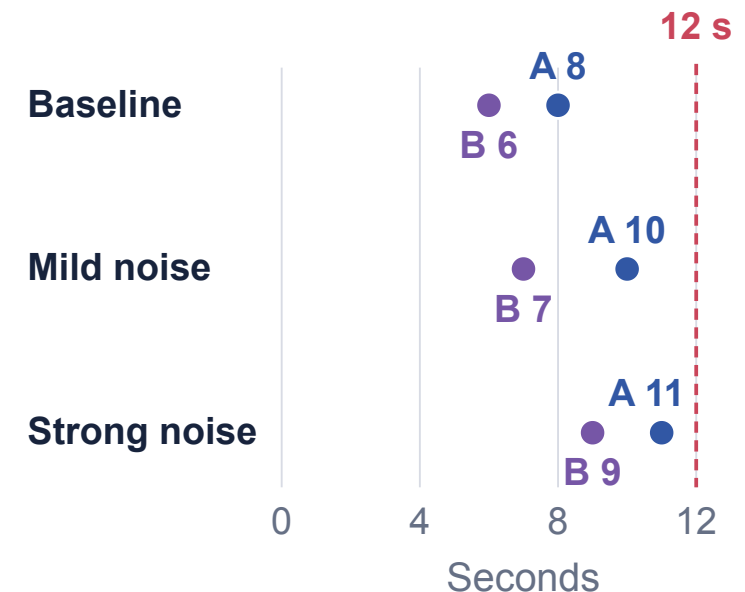


Left + larger: primary judgment

More space emphasizes the main claim.

Maximum observed runtime

Maximum of 8 recorded runs



Right + smaller: supporting judgment

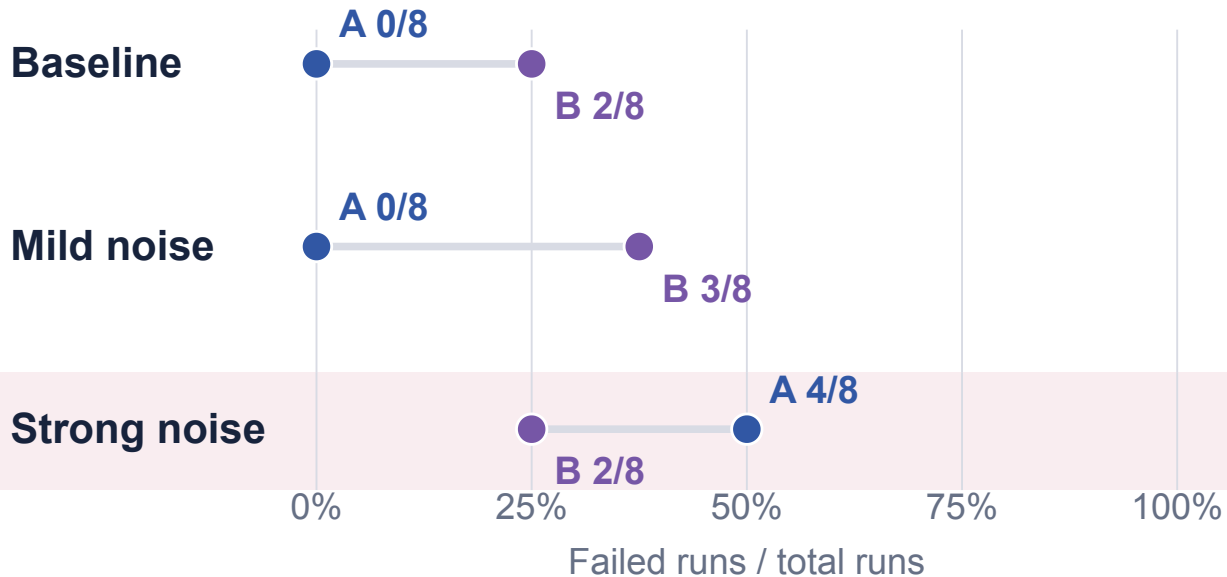
Checks the extra 12-second limit.

Complementarity adds a required judgment

Beyond reliability, the task requires every run to finish within **12 seconds**.

Observed failures across conditions

Failure: score < 70 · 8 runs per method / condition



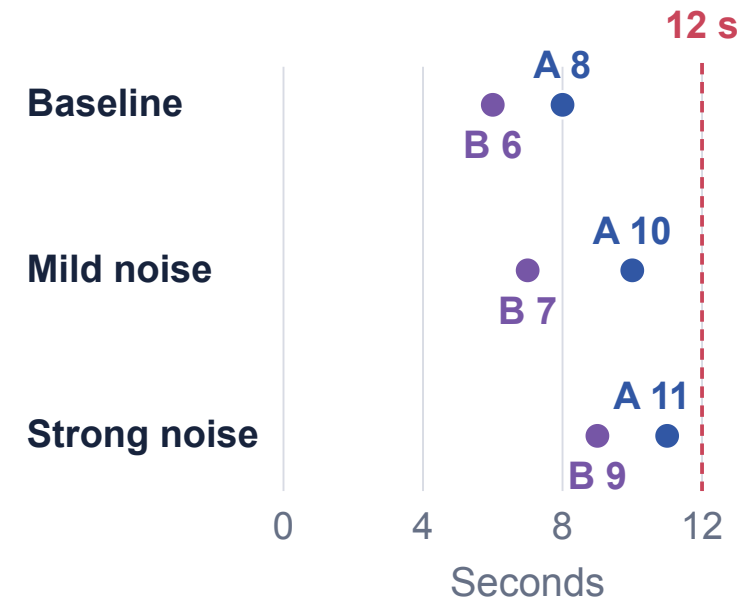
Main view: reliability

Which method fails less often in each condition?

Cannot answer: do runs meet the time limit?

Maximum observed runtime

Maximum of 8 recorded runs

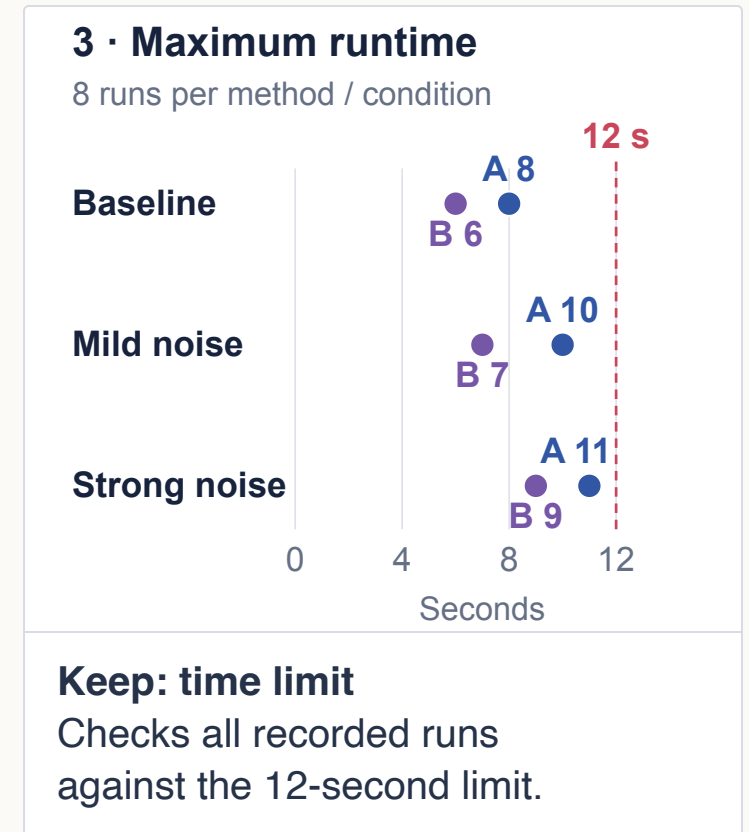
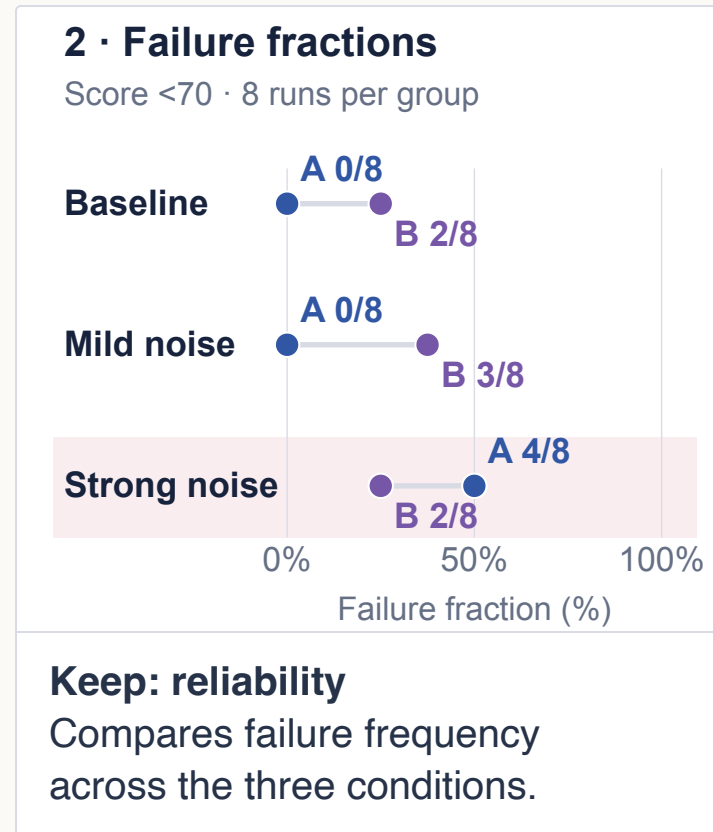
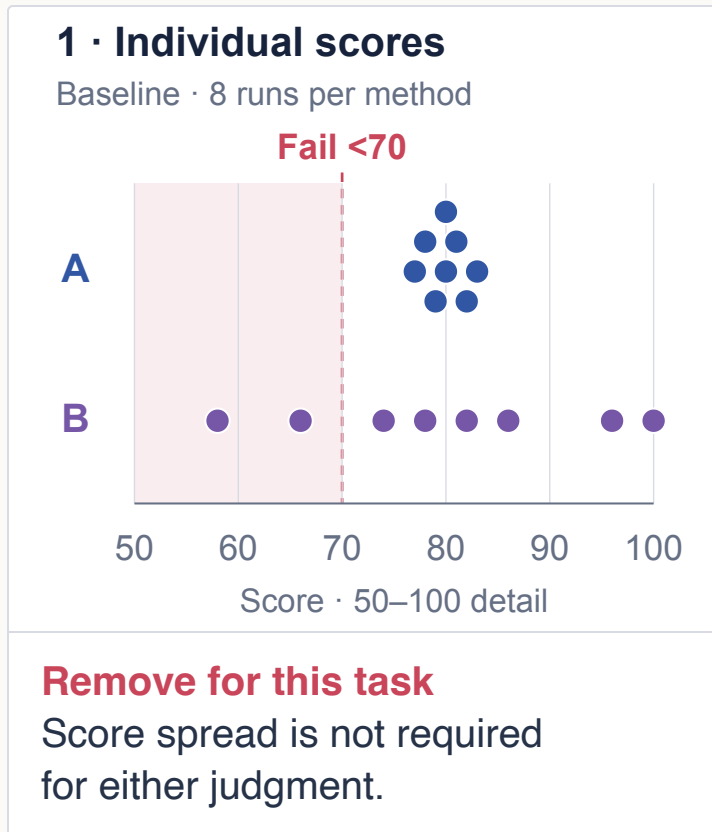


Added view: runtime

Maximum runtime below 12 seconds shows that every recorded run meets the time limit.

Necessity tests whether a whole view is needed

Task: Show "A fails less often than B across the noise conditions" and check the 12s limit.



Removal test: without View 1, Views 2 and 3 still support both required judgments.

Coordinate views for comparison

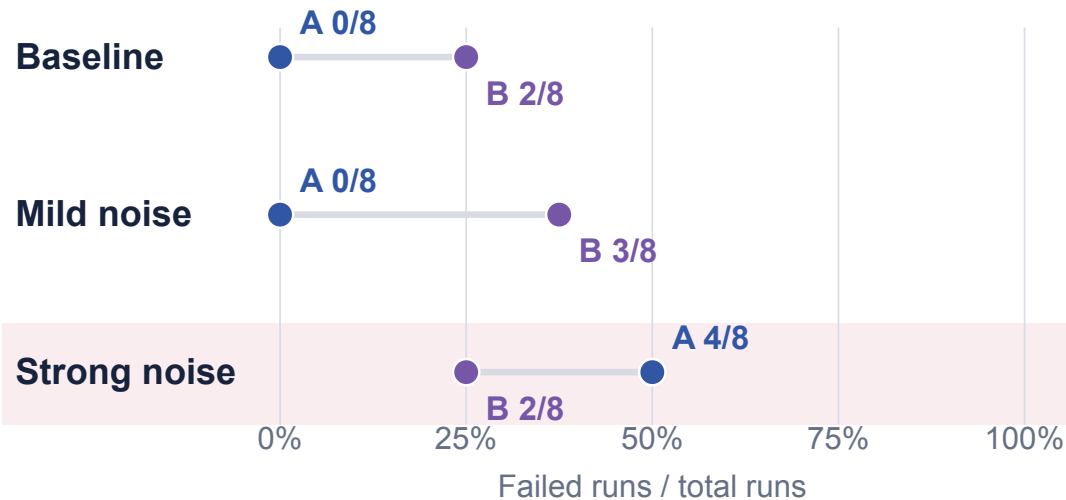
Across views, readers need to match the same objects and know when visual magnitudes can be compared.

- **Alignment** places corresponding elements in different views on common rows or columns.
Keep baseline, mild noise, and strong noise in the same order and align their rows.
- **Shared encodings** use the same visual channel and value mapping for a common variable.
Method A stays blue and Method B stays violet in both views.
- **Scale choice** controls how data values map to positions within each view.
Compare failure rates on a shared scale; keep seconds on a separate scale.

Aligned rows connect the same condition across views

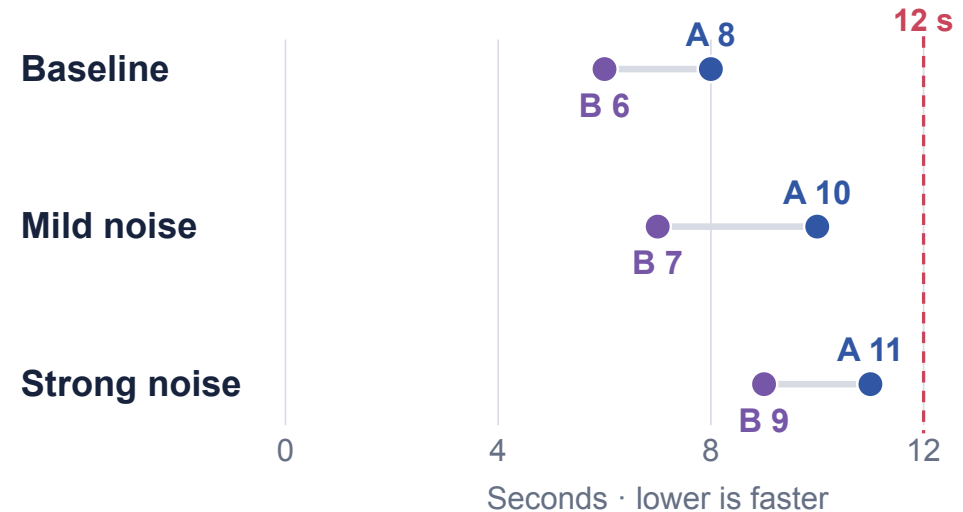
Observed failures across conditions

Failure: score <70 · 8 runs per method / condition



Maximum observed runtime

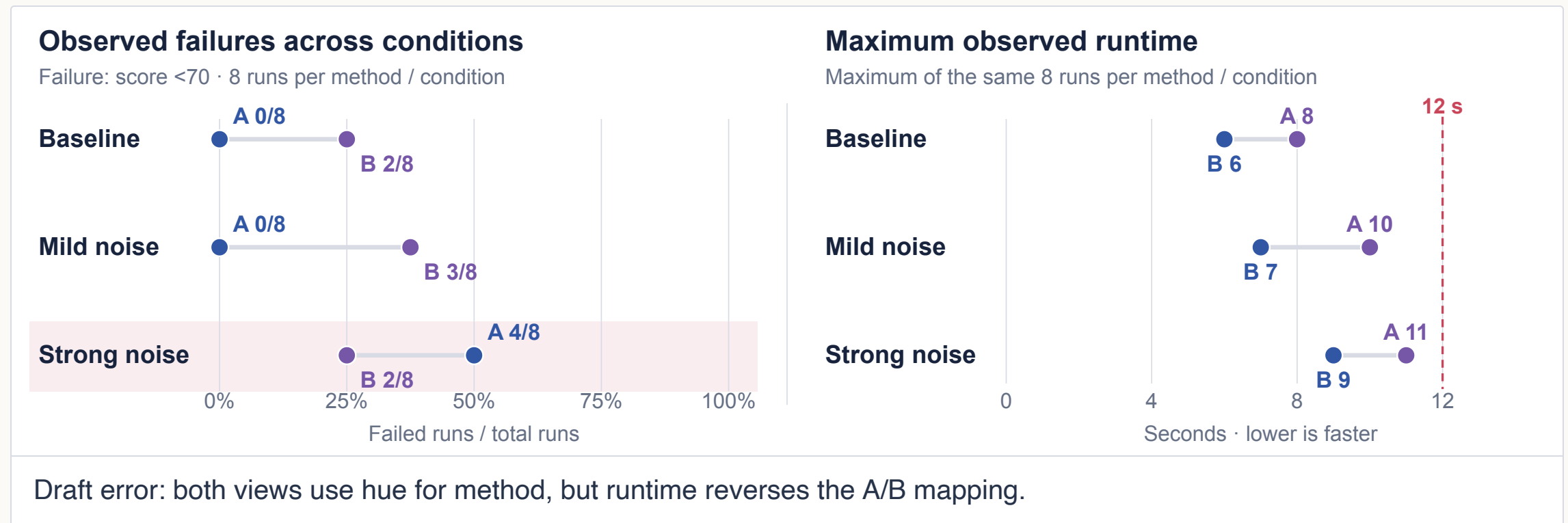
Maximum of the same 8 runs per method / condition



Aligned rows match the same condition across views. Failure rate and runtime retain separate scales.

Shared encodings preserve identity across views

Shared encodings use the same visual channel and value mapping for a variable that appears in multiple views.



Shared mapping: Method A → blue; Method B → violet in both views.

Choose scales for the comparison readers must make

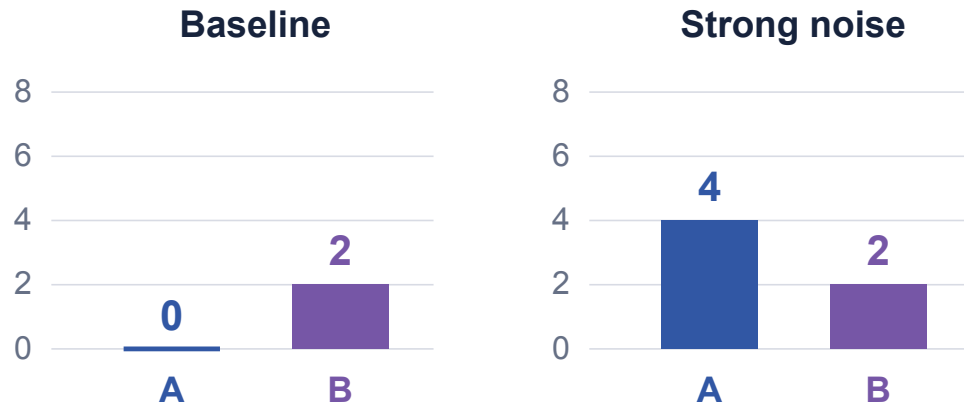
Shared scales use the same mapping from data values to visual positions across comparable views. **Independent scales** give each view its own mapping.

- Share scales when readers must compare **magnitudes of the same quantity**.
- Independent scales can expose local detail, but cross-panel heights or distances may no longer be comparable.
- Different units, such as failure rate and seconds, require clearly labeled, separate scales.

Shared scales support direct comparison across views

Shared count scale

Same data · failure: score <70 · 8 runs each



B = 2 in both conditions; compare the axes.

Shared scales: B's 2 failures have the same bar height in both conditions.

Independent count scales

Same data · failure: score <70 · 8 runs each

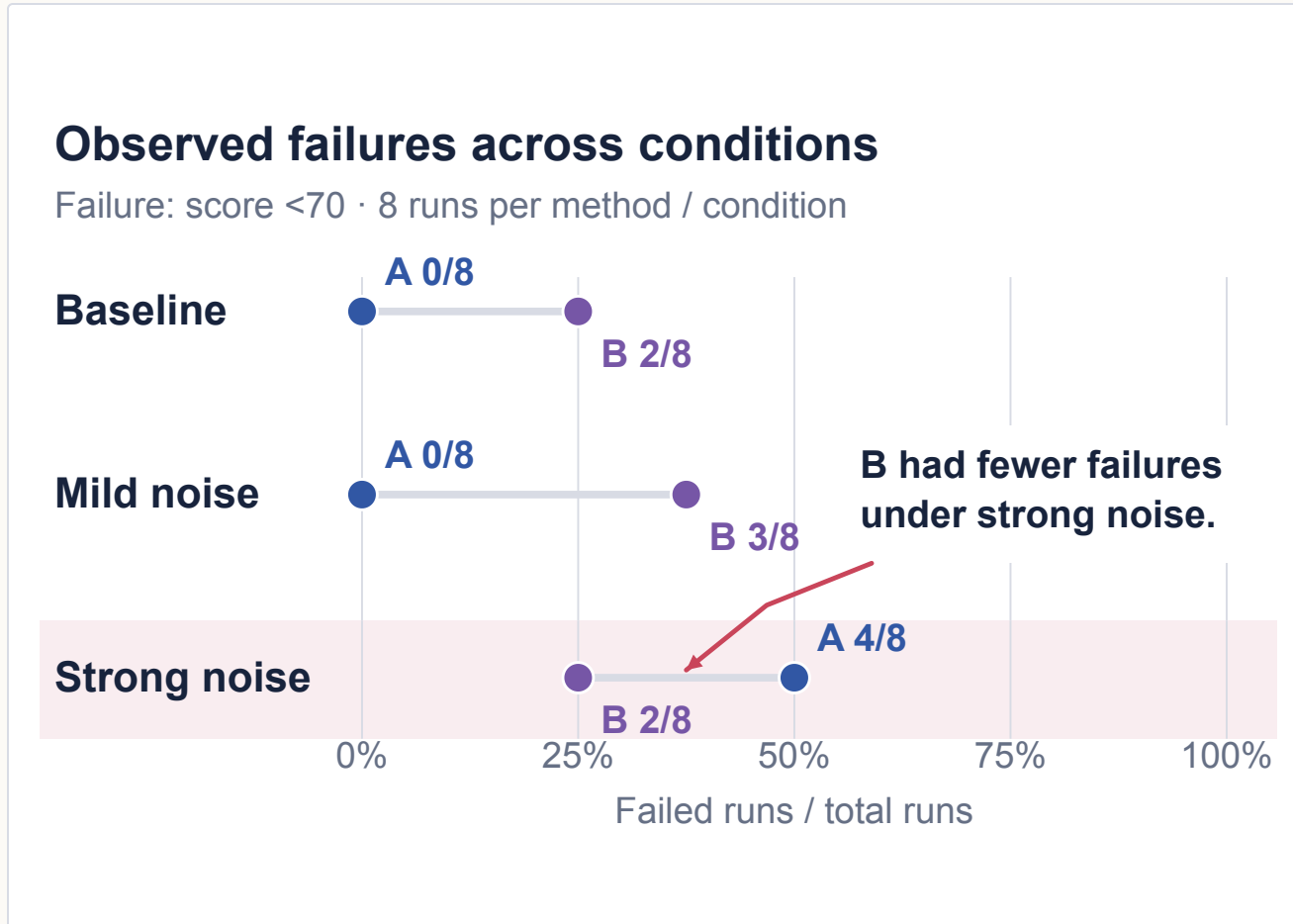


B = 2 in both conditions; compare the axes.

Independent scales: B's 2 failures have different bar heights across conditions.

For comparing failure frequency across conditions, preserve a common scale.

Labels and annotations clarify comparisons



Labels identify the evidence

Use the same method and condition names across views. Keep units, reference values, and count labels beside the relevant marks.

Annotation states an observation

The text and leader line connect the observation to the strong-noise comparison.

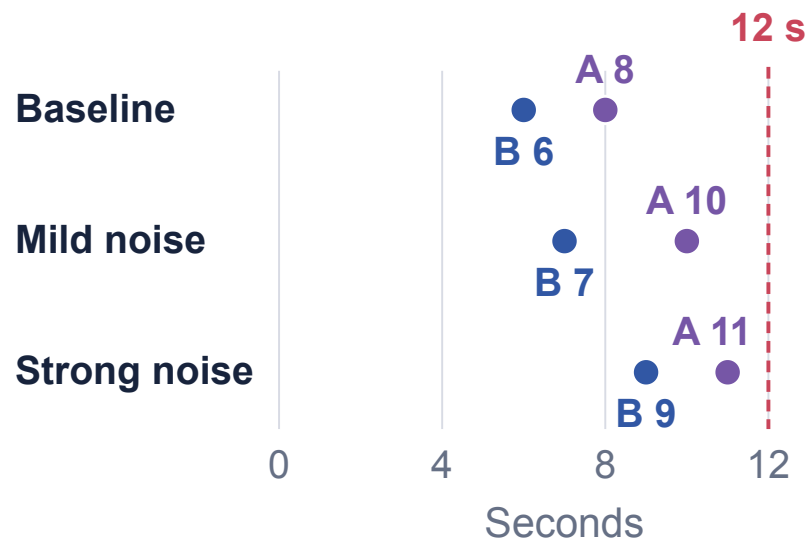
A useful annotation directs attention without exceeding the evidence.

Practice: repair the reliability figure as a whole

DRAFT · diagnose the visual decisions

Maximum observed runtime

Maximum of 8 recorded runs



Failed counts

Independent axes · each out of 8

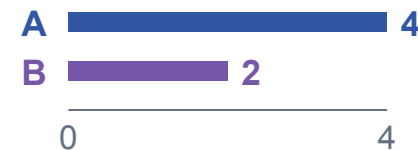
Baseline



Mild noise



Strong noise



Failed percentages

The same failures again

Baseline

A 0% B 25%

Mild noise

A 0% B 37.5%

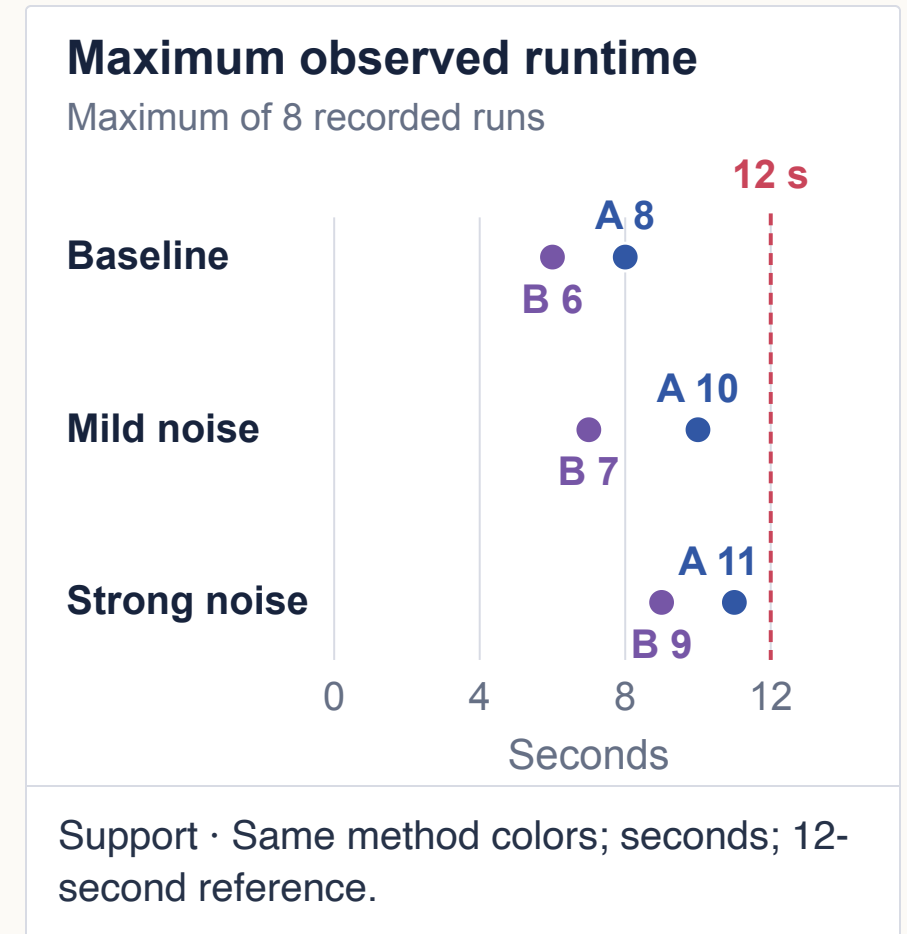
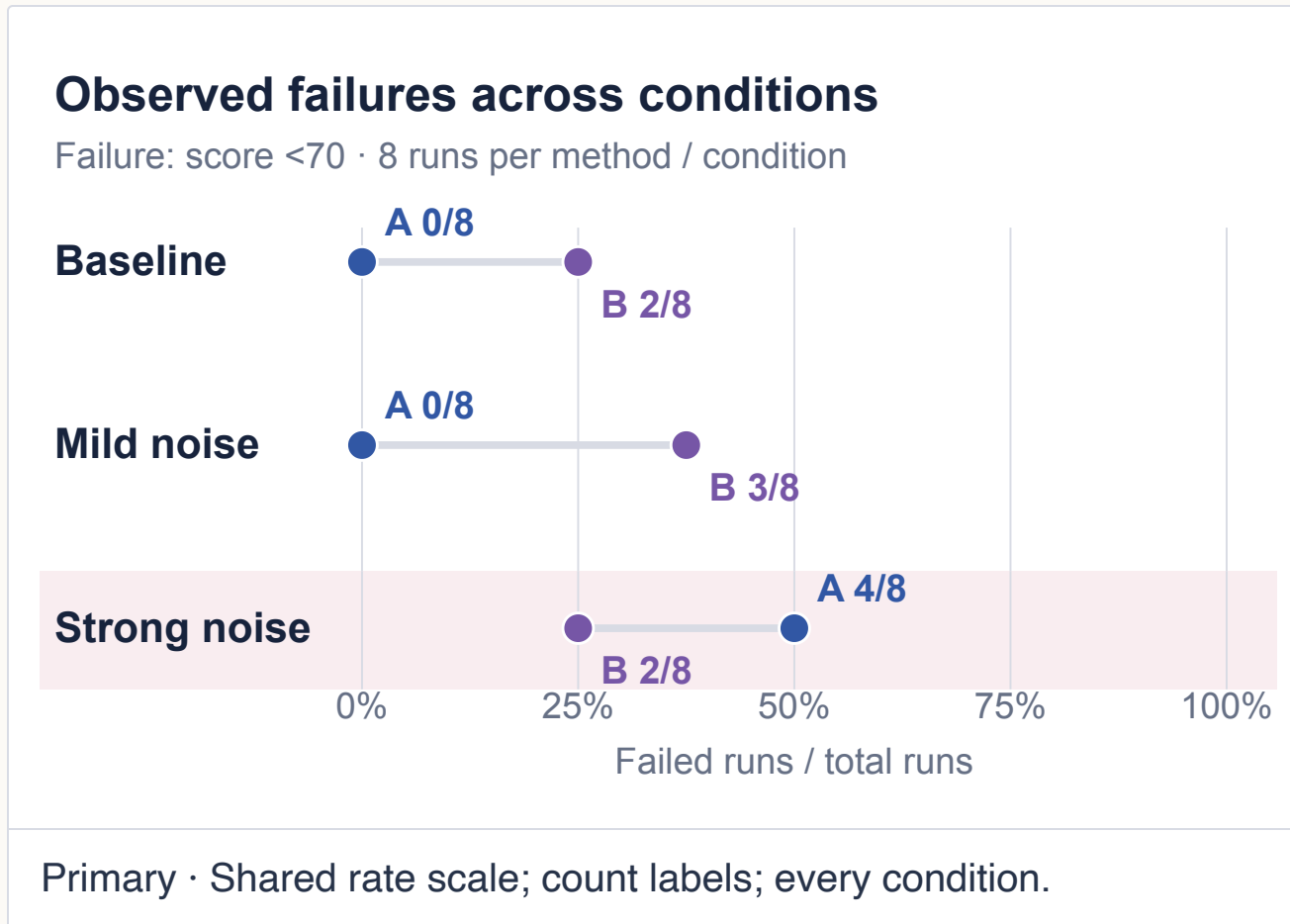
Strong noise

A 50% B 25%

Brief: Compare failures across conditions, then check runtime. Sketch a revision and explain **three design changes**.

Check reading order, view necessity, and comparisons across views.

Reference: make the primary comparison easy to inspect



The figure is easier to read. Is the evidence sufficient for the conclusion?

PART 04

Figure integrity

How can design preserve a justified interpretation?

Four design principles preserve figure integrity

Figure integrity is the preservation of the information readers need to interpret displayed evidence in a scientifically justified way.

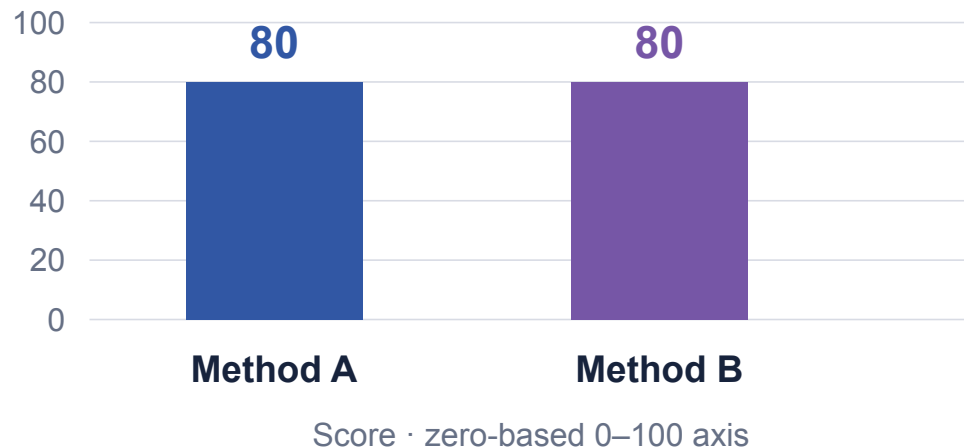
- **Preserve the evidence needed for the judgment.**
Choose summaries that keep relevant comparisons and exceptions.
- **Represent variability and uncertainty appropriately.**
Show outcome spread or estimate precision, and label what the marks mean.
- **Make analytical choices explicit.**
State rules, denominators, and transformations that affect interpretation.
- **Keep conclusions within the evidence.**
Name the tested conditions and distinguish recorded outcomes from broader claims.

Preserve the evidence needed for the judgment

A **summary** describes a property of the observations. Choose one that preserves the comparison required by the claim.

Baseline · mean score

8 runs per method · Average score



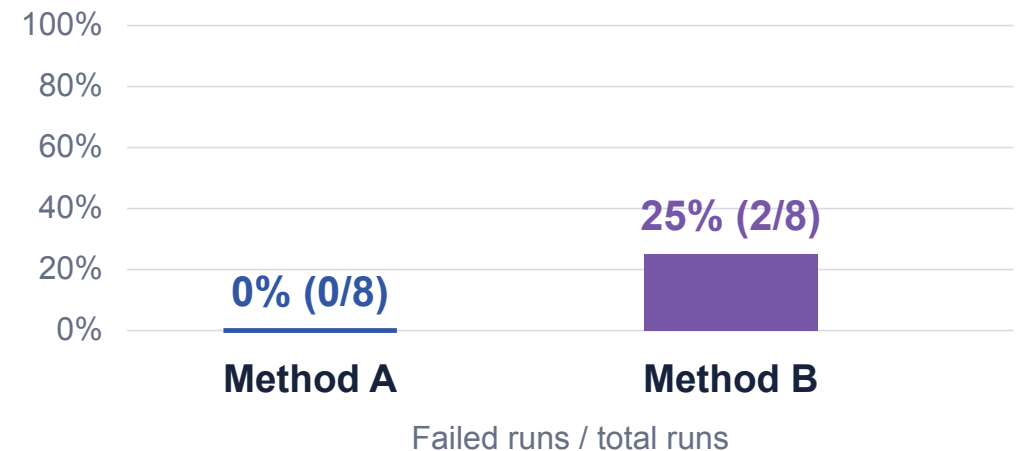
Claim: “A and B have the same mean score.”

Compare averages → use **mean score**.

In these baseline runs: both means are 80.

Baseline · failure fraction

Failure: score <70 · 8 runs per method



Claim: “A fails less often than B.”

Compare failure frequency → use **failure fraction**.

In these baseline runs: A 0/8; B 2/8.

Choose how to show variability and uncertainty

Variability is the spread of observed outcomes. **Estimate uncertainty** describes how much an estimated mean could change if we repeated the study with a new sample.

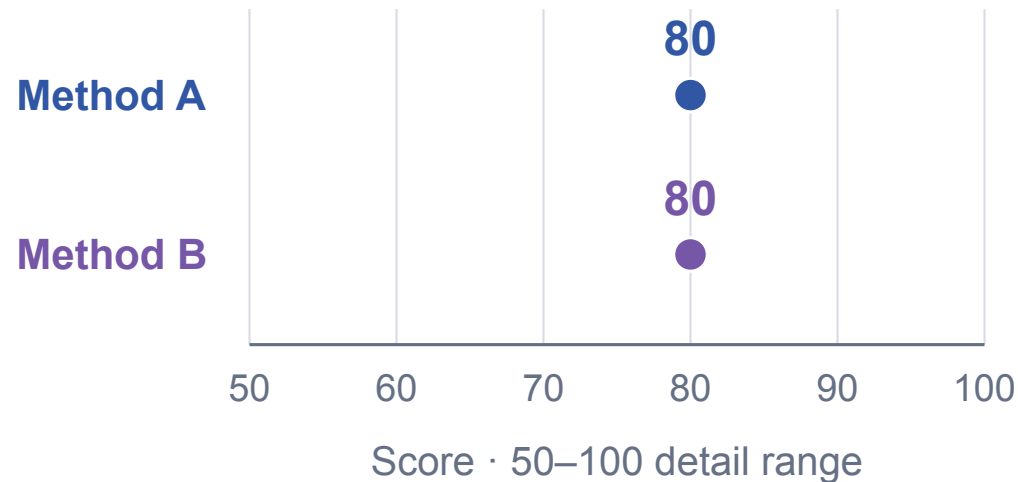
To show	Design action
Outcome variability	Show individual values or a distribution in the outcome's units.
Estimate uncertainty	Pair the estimate with an appropriate uncertainty interval for that quantity.

Show variability alongside the summary

When to use it: When readers need to judge how much individual outcomes vary, show the observations or their distribution.

Baseline · mean score

8 runs per method · Each point is a mean



Mean only: both methods average 80.
The spread across runs is not shown.

Baseline · individual scores

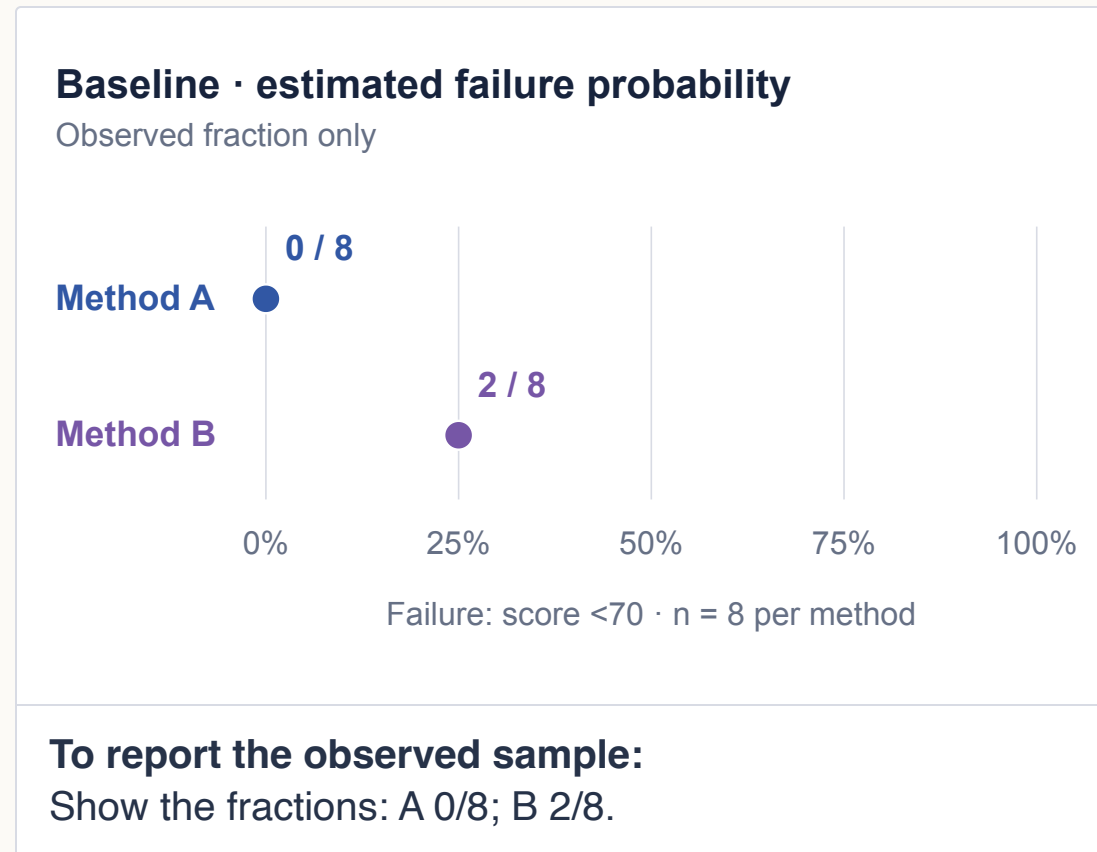
8 runs per method · Each dot is one run



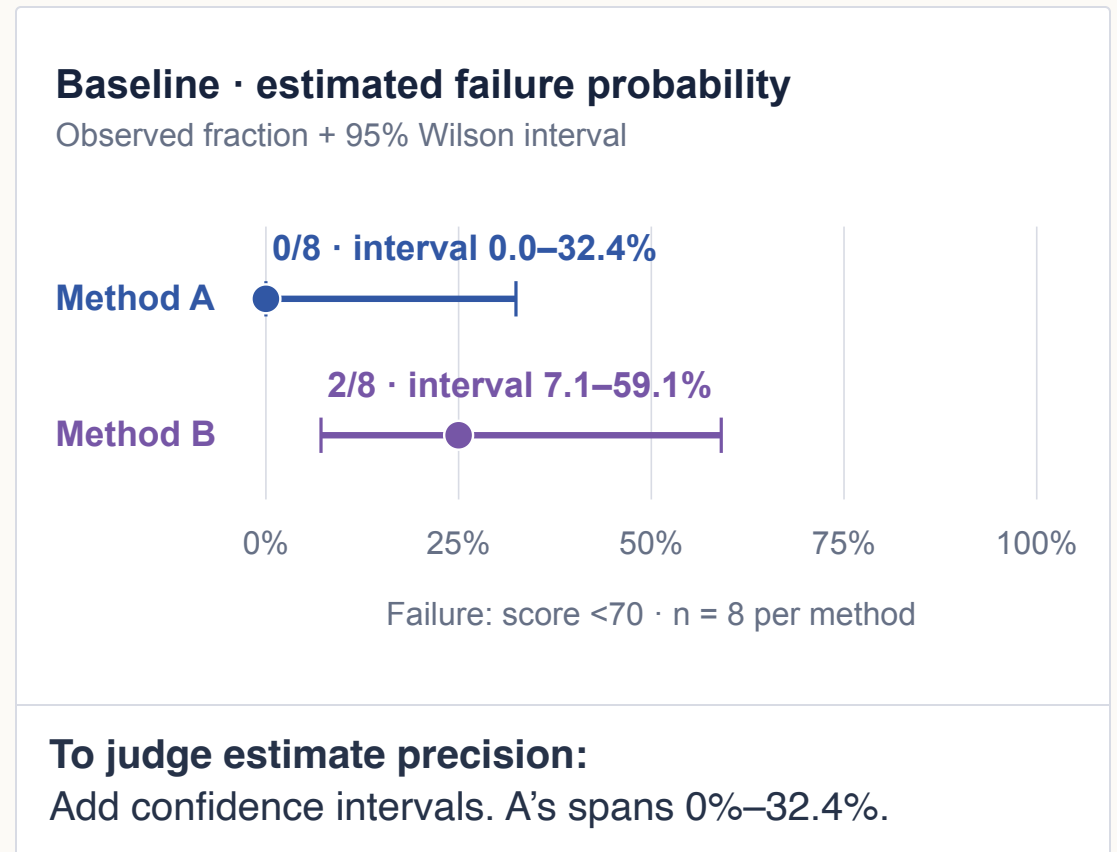
Individual runs: A 77–83; B 58–100.
B's recorded scores vary more.

Show uncertainty alongside the estimate

When to use it: When using a sample to estimate an underlying mean or failure probability, show an interval so readers can judge precision.

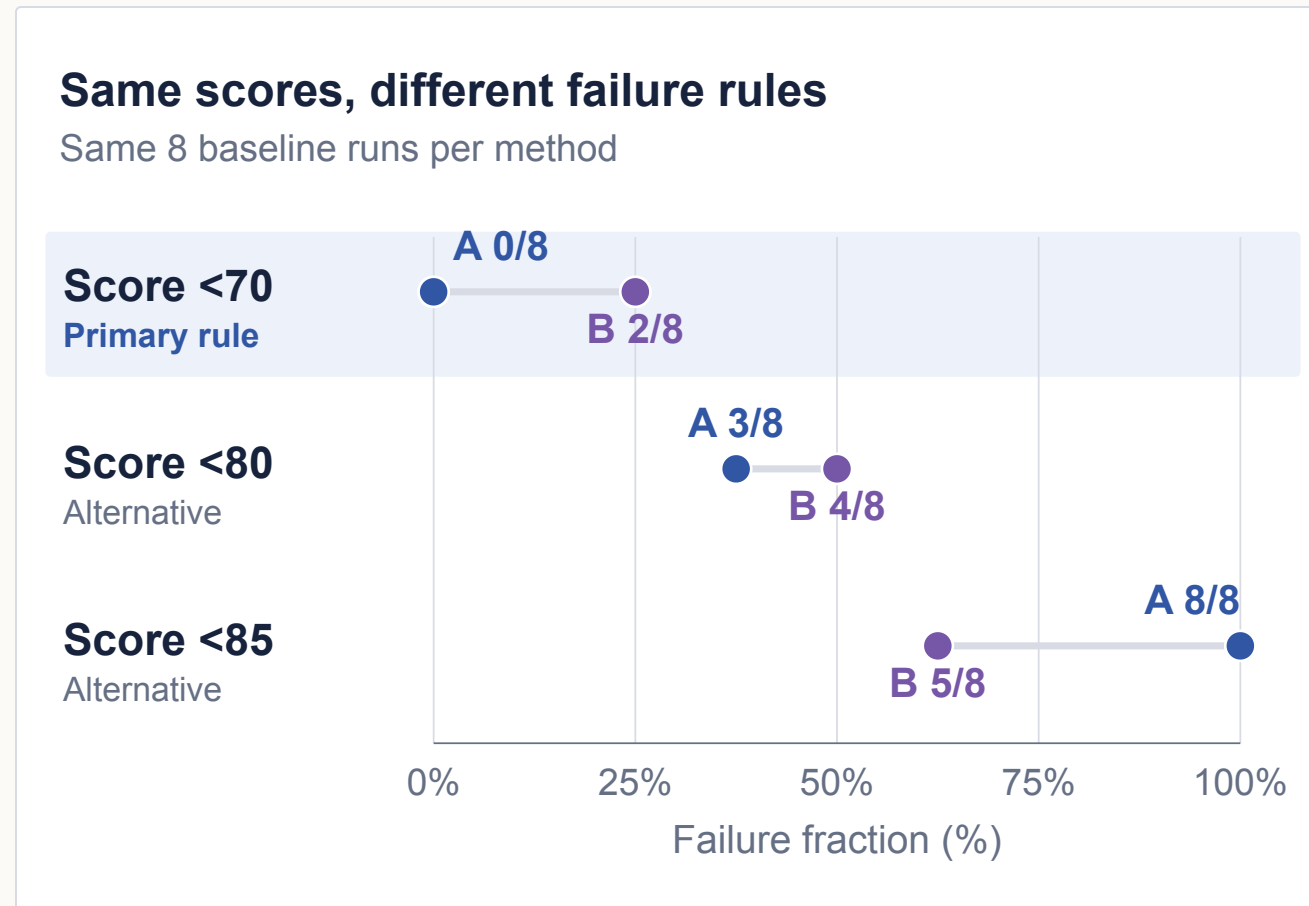


n = 8 per method · Independent binomial runs · 95% Wilson intervals · Method: NIST.



Make analytical choices visible in the figure

When to use it: If different defensible rules could change the conclusion, compare them and label the rule used in each case.



The cutoff changes the answer

A has fewer failures below 70; B has fewer below 85. The scores are the same.

Make that dependence visible

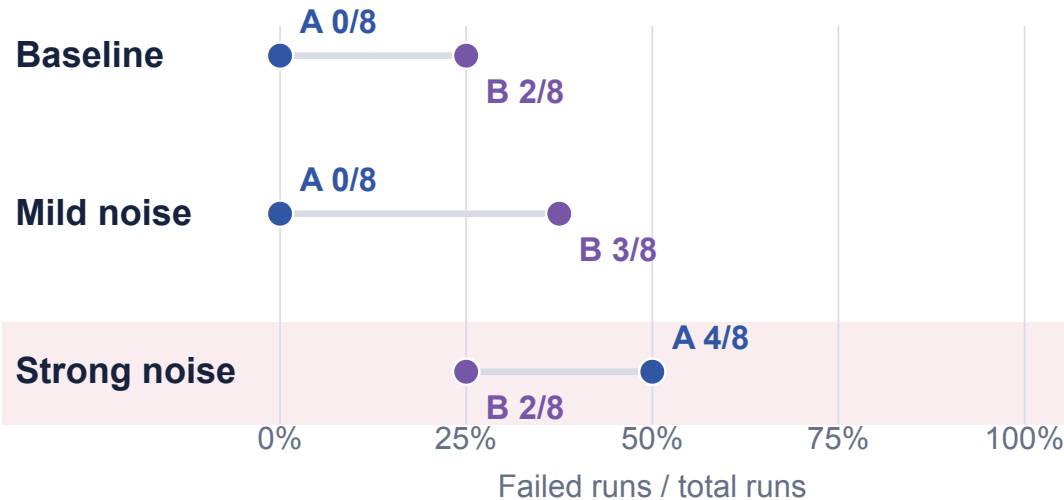
- **Highlight the primary rule:** score <70.
- **Compare alternative cutoffs:** use labeled rows on the same scale.
- **Keep the denominator visible:** failures out of 8 runs.

Keep conclusions within the displayed evidence

When to use it: If a broad claim such as “A consistently outperforms B” goes beyond the displayed results, narrow it to the tested conditions and keep exceptions.

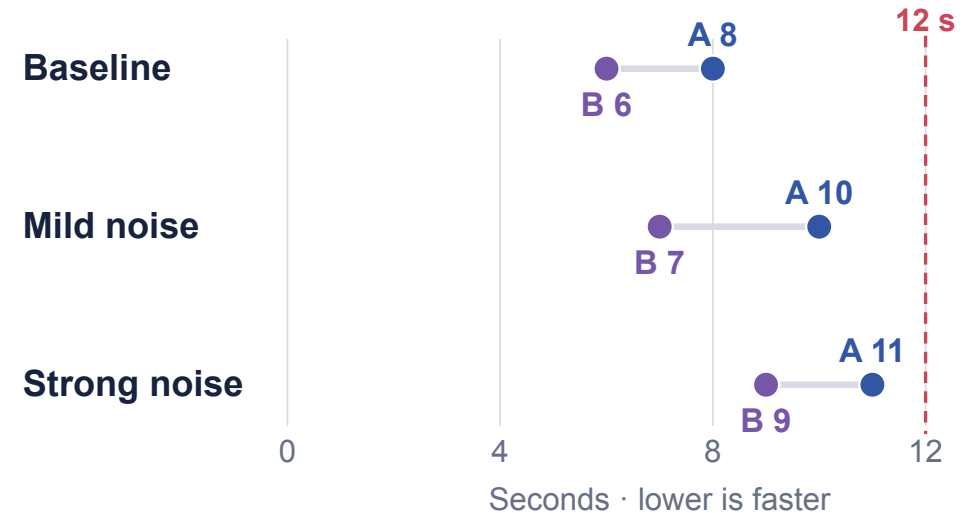
Observed failures across conditions

Failure: score <70 · 8 runs per method / condition



Maximum observed runtime

Maximum of the same 8 runs per method / condition



Among the **8 recorded runs per method in each tested condition**, A had fewer failures at baseline and mild noise, while **B had fewer under strong noise**. All recorded runtimes were below 12 s.

From required judgments to inspectable evidence

1. **Claim to comparison:** define failure, then compare A and B in each condition.
2. **One view or multiple views:** share a view for the failure judgments; add a view for runtime.
3. **Designing a multi-view figure:** prioritize failures, retain runtime, and coordinate views for comparison.
4. **Integrity:** preserve relevant evidence, represent uncertainty, state analytical choices, and bound the conclusion.

The resulting finding is bounded: A had fewer failures at baseline and mild noise; B had fewer under strong noise.

A scientific evidence figure lets readers inspect both the result and its limits.